$$\left\langle \hat{\sigma}_z^{(i)}(\tau)\hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_\mathcal{E}} = \left\langle \hat{\sigma}_z^{(i)}(\tau) \right\rangle_{\hat{\rho}_\mathcal{E}} \left\langle \hat{\sigma}_z^{(j)} \right\rangle_{\hat{\rho}_\mathcal{E}} \tag{5.173}$$